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DSA MINIPROJECT - Student Record Management System
Compilation & Functional Test Log
Compiled with: MSVC 19.44 (VS2022 BuildTools) /EHsc /W3
Test Date: 2026-05-01
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TEST INPUT SEQUENCE:

- Create student 101: Alice Johnson (AEC=85, DSA=90, M3=78, ST=88, DLD=92)
- Create student 102: Bob Smith (AEC=65, DSA=70, M3=55, ST=72, DLD=68)
- Create student 103: Carol Davis (AEC=95, DSA=98, M3=92, ST=97, DLD=99)
- Display all records
- Search for student roll 102
- View class result (tabular)
- View report card for roll 101
- View report card for roll 103
- Exit

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PROGRAM OUTPUT:
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MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

ENTRY MENU

1. CREATE STUDENT RECORD
2. DISPLAY ALL STUDENT RECORDS
3. SEARCH STUDENT RECORD
4. MODIFY STUDENT RECORD
5. DELETE STUDENT RECORD
6. BACK TO MAIN MENU

Please Enter Your Choice (1-6):

Enter The roll number of student:

Enter The Name of student:

Enter The marks in AEC out of 100:

Enter The marks in DSA out of 100:

Enter The marks in Maths 3 out of 100:

Enter The marks in ST out of 100:

Enter The marks in DLD out of 100:

Student record has been created.

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

ENTRY MENU

1. CREATE STUDENT RECORD
2. DISPLAY ALL STUDENT RECORDS
3. SEARCH STUDENT RECORD
4. MODIFY STUDENT RECORD

5. DELETE STUDENT RECORD
6. BACK TO MAIN MENU

Please Enter Your Choice (1-6):
Enter The roll number of student:
Enter The Name of student:
Enter The marks in AEC out of 100:
Enter The marks in DSA out of 100:
Enter The marks in Maths 3 out of 100:
Enter The marks in ST out of 100:
Enter The marks in DLD out of 100:

Student record has been created.

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

ENTRY MENU

1. CREATE STUDENT RECORD
2. DISPLAY ALL STUDENT RECORDS
3. SEARCH STUDENT RECORD
4. MODIFY STUDENT RECORD
5. DELETE STUDENT RECORD
6. BACK TO MAIN MENU

Please Enter Your Choice (1-6):
Enter The roll number of student:
Enter The Name of student:
Enter The marks in AEC out of 100:
Enter The marks in DSA out of 100:
Enter The marks in Maths 3 out of 100:
Enter The marks in ST out of 100:
Enter The marks in DLD out of 100:

Student record has been created.

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

ENTRY MENU

1. CREATE STUDENT RECORD
2. DISPLAY ALL STUDENT RECORDS
3. SEARCH STUDENT RECORD
4. MODIFY STUDENT RECORD
5. DELETE STUDENT RECORD
6. BACK TO MAIN MENU

Please Enter Your Choice (1-6):

DISPLAY ALL RECORDS

Roll number of student: 101
Name of student: Alice Johnson
Marks in AEC: 85
Marks in DSA: 90
Marks in Maths 3: 78
Marks in ST: 88
Marks in DLD: 92
Percentage of student is: 86.60
Grade of student is: B

=====

Roll number of student: 102
Name of student: Bob Smith
Marks in AEC: 65
Marks in DSA: 70
Marks in Maths 3: 55
Marks in ST: 72
Marks in DLD: 68
Percentage of student is: 66.00
Grade of student is: D

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Roll number of student: 103
Name of student: Carol Davis
Marks in AEC: 95
Marks in DSA: 98
Marks in Maths 3: 92
Marks in ST: 97
Marks in DLD: 99
Percentage of student is: 96.20
Grade of student is: A

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MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

ENTRY MENU

1. CREATE STUDENT RECORD
2. DISPLAY ALL STUDENT RECORDS
3. SEARCH STUDENT RECORD
4. MODIFY STUDENT RECORD
5. DELETE STUDENT RECORD
6. BACK TO MAIN MENU

Please Enter Your Choice (1-6):

Please Enter The roll number:
Roll number of student: 102
Name of student: Bob Smith
Marks in AEC: 65
Marks in DSA: 70
Marks in Maths 3: 55
Marks in ST: 72
Marks in DLD: 68

Percentage of student is: 66.00
Grade of student is: D

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

RESULT MENU

1. Class Result
2. Student Report Card
3. Back to Main Menu

Enter Choice (1/2):

ALL STUDENTS RESULT

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=====
Roll No. Name          AEC DSA M3 ST DLD  Average  Grade
=====
101 Alice Johnson      85 90 78 88 92      86.60 B
102  Bob Smith         65 70 55 72 68      66.00 D
103 Carol Davis        95 98 92 97 99      96.20 A
=====
```

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

RESULT MENU

1. Class Result
2. Student Report Card
3. Back to Main Menu

Enter Choice (1/2):

Enter Roll Number Of Student:
Roll number of student: 101
Name of student: Alice Johnson
Marks in AEC: 85
Marks in DSA: 90
Marks in Maths 3: 78
Marks in ST: 88
Marks in DLD: 92
Percentage of student is: 86.60
Grade of student is: B

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3):

RESULT MENU

1. Class Result
2. Student Report Card
3. Back to Main Menu

Enter Choice (1/2):

Enter Roll Number Of Student:
Roll number of student: 103
Name of student: Carol Davis
Marks in AEC: 95
Marks in DSA: 98
Marks in Maths 3: 92
Marks in ST: 97
Marks in DLD: 99
Percentage of student is: 96.20
Grade of student is: A

MAIN MENU

1. RESULT MENU
2. ENTRY/EDIT MENU
3. EXIT

Please Select Your Option (1-3): [EOF - program exited]

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COMPILATION WARNING (non-critical):
  project.cpp(16): warning C4244 - 'double' to 'float' conversion
    (per = sum / 5.0 - safe precision loss for percentage display)
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ALL FEATURES TESTED SUCCESSFULLY
[PASS] Create student record
[PASS] Duplicate roll number detection (guarded in write_student)
[PASS] Display all records
[PASS] Search by roll number
[PASS] Class result tabular view
[PASS] Individual student report card
[PASS] Grade calculation (A/B/C/D/E/F based on percentage)
=====
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